Technology Stack

Date	30 September 2026
Team ID	SWTID-2026-3818
Project Name	Auto_Ticket_Classification_Using_Flow_Diagram
Maximum Marks	2 Marks

The table lists the technologies chosen for each layer of the architecture, with a short justification.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Jinja2 templates	Lightweight, works in any browser, easy to connect with Flask templates.
2	Backend / Server-Side	Python 3.10+, Flask	Python is the standard language for ML; Flask is simple and quick to build REST endpoints.
3	Machine Learning / NLP	scikit-learn (TF-IDF, Logistic Regression / Linear SVM), NLTK, pandas, NumPy, joblib	Proven libraries for text classification; the model is easy to train, save and reload.
4	Database / Data Storage	SQLite (CSV files for training data)	No server setup needed, ideal for a student project; can be moved to PostgreSQL later.
5	Flow Diagram Generation	Graphviz (python-graphviz)	Creates clear flow diagrams automatically from code for each ticket.
6	Data Visualisation	matplotlib, seaborn (evaluation) and Chart.js (dashboard)	Confusion matrix and accuracy charts for evaluation; interactive charts for the admin dashboard.
7	Cloud / Hosting / Deployment	Local Flask server; optional Ngrok or Render for a public link	Free and quick to demonstrate; no infrastructure cost.
8	Version Control & CI/CD	Git and GitHub	Team collaboration and tracking of changes.
9	Development Environment	VS Code, Jupyter Notebook / Google Colab	Notebook for experiments, VS Code for the application code.
10	Testing Tools	Postman, Locust, pytest	API testing, load testing and unit testing.